Seeing History Is Not Learning to Use It: An Exploratory Task-Inference Pilot

Abstract

An interactive policy must exploit evidence about an unknown task, not merely receive a long history. We investigate this distinction in the original DREAM Cooking environment using three recurrent behavior-cloning objectives with matched data, model size, initialization and optimizer updates. Over three optimizer seeds, observed-fact supervision achieves 56.80% success, compared with 24.21% for action-only supervision and 34.29% for current-state auxiliary supervision. Resetting task memory reduces factual-model success to 24.00%; a separately registered feedback-redaction diagnostic reduces it to 29.81%. Independent replay verifies all actions and outcomes, while a symbolic full-history reference remains at 100%. These are exploratory representation-learning results, not a novel auxiliary algorithm, on-policy distillation, robot result or conference-readiness claim.

1. Research Question and Prior Art

Tasks can share observations and control skills while differing in an unknown action–outcome mapping. Prior interaction may reveal the mapping even when it does not immediately finish the requested task. Our broader question is how experience should train the policy's task-inference process, in addition to its next action. A necessary first check is whether ordinary action supervision learns to exploit an informative history.

This question has substantial prior art. DREAM separates task-relevant exploration from exploitation (Liu et al., 2021). Task-belief supervision, including auxiliary prediction heads, was investigated by Humplik et al. (2019); VariBAD learns approximate inference and conditions behavior on uncertainty (Zintgraf et al., 2020). UNREAL already demonstrates representation benefits from auxiliary objectives (Jaderberg et al., 2017). Consequently, neither a factual prediction head nor the distinction between acquiring and using knowledge is claimed as an invention here. The pilot isolates a useful diagnostic before pursuing a narrower experience-update mechanism.

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2. Learning Object and Observable Boundary

Let τ index an unknown drawer mapping, and H_t denote the observed interaction prefix across attempts on that mapping. The recurrent policy state satisfies

$$m_t = U_\theta(m_{t-1}, x_t), \qquad \pi_\theta(a_t \mid m_t), \tag{1}$$

where x_t contains visible observation, revealed goal, previous actual action and reward, episode boundary, phase and local time. The mapping ID, hidden assignment, factual labels, teacher and symbolic memory are excluded from inference. State is retained across attempts on one task and cleared at the next task. An actual terminal observation and reward are ingested before each reset as a feedback-only token; no extra environment action is spent.

All arms use a 96-unit embedding and 96-unit causal GRU with identical action and auxiliary head sizes (63,576 parameters). The common training objective is

$$\mathcal{L}_{\text{action}} + 0.2\mathcal{L}_{\text{aux}}, \tag{2}$$

with the auxiliary term zero for action-only training. The current-state control predicts inventory and the first two visible bowl slots. The factual target reconstructs the latest observed food in each of three drawers, leaving an unobserved drawer unknown. Targets are generated exclusively from actual pickup observations at or before the input prefix. They are never fed to the policy.

3. Protocol and Comparability

We pin the official DREAM implementation to revision `d52204c` (full identity in the plan), preserving transition, reward, observation and the 20-step horizon. Our explicitly registered 32/8/24 mapping split differs from the original paper's setup and reuses our earlier diagnostic panel. This is not a DREAM performance reproduction or an untouched confirmatory test.

The common training set has 1,062 histories over 32 mappings and 177 valid goals: three behavior seeds, each with deterministic and fixed 10% action-perturbation collection.

Table 1. Five-attempt mean success (%). Three optimizer seeds are reported completely. Reset uses the same checkpoint with task memory cleared between episodes.

Objective	Mean	Seed range	Reset
Action-only	24.21	20.00–27.84	18.67
Current-state aux.	34.29	31.84–36.48	18.13
Observed-fact aux.	56.80	51.68–60.80	24.00
Symbolic reference	100.00	100.00–100.00	–

The teacher acts only on observed facts. This provides limited recovery-state coverage, not student-on-policy data. For each optimizer seed (7, 19, 41), the three arms share initial weights and minibatch indices. Each receives 800 Adam updates at learning rate 0.003, batch size 24 sequences and gradient clipping 1. The final checkpoint is fixed; development does not select it. Nine runs spend 7,200 optimizer updates.

Each evaluated task includes up to 20 exploration actions followed by five executions of the same goal. The learned policy controls from the initial step. The 24 evaluation mappings provide 125 valid mapping–goal cases; each arm contributes 1,875 execution episodes across three checkpoints. Repeated episodes are nested under mapping and optimizer seed, not independent training replicates. A failed execution receives cost 20; actual action counts are also recorded.

4. Results and Separating Diagnostics

Factual supervision exceeds action-only by 32.59 percentage points and the current-state control by 22.51 points, with positive paired differences in all three seeds. Memory reset causes a 32.80-point drop. These meet the registered pilot predictions (five-point differences, positive direction per seed and a five-point memory effect). Three seeds support descriptive comparisons, not a general superiority test. High teacher-forced training accuracy is not used as closed-loop efficacy evidence.

Factual success rises only from 54.93% on the first execution to 57.60% on the fifth. The first execution already follows 20 interaction steps and is not zero-shot. The strongest evidence concerns using acquired information, rather than progressively better experimentation or recursive self-improvement.

Shared history and feedback use. After reading the primary results, we register a second experiment using the nine frozen checkpoints. Each receives a common real 20-step exploration prefix. The paired intervention replaces only prefix inventory observations with EMPTY at learner input; execution observations remain unchanged. Raw logs retain original observations. This supplied-prefix diagnostic is distinct from primary teacher-free exploration and uses no gradients.

Independent comparison finds that all 1,125 autonomous prefixes already equal the supplied common prefix. Thus, differences in exploration information do not explain the primary comparison on these cases. Redaction collapses 125 prefix input sequences to one. The complete results are: The factual model uses historical feedback; merely possessing a recurrent history is insufficient for the action-only baseline. Redaction slightly improves action-only success and changes the input distribution. We therefore do not interpret the intervention as universal benefit from memory or as proof of a new causal-credit theory.

Table 2. Shared-prefix feedback intervention (%). Registration occurred after the primary pilot.

Model	Real	Redacted	Difference
Action-only	24.21	26.99	−2.77
Current-state aux.	34.29	32.91	+1.39
Observed-fact aux.	56.80	29.81	+26.99

5. Integrity, Costs and Failure History

The first registered learner run omitted terminal feedback from the GRU despite logging it. This could let the next episode's auxiliary target refer to unseen evidence. We interrupted that run before evaluation, retained its code and plan, and registered a correction adding actual feedback-only tokens. Task, split, horizon, model, loss weight, seeds and update budget did not change. No scientific metric uses that run. Its overhead is 79,904 source actions and 2,700–2,799 optimizer updates.

The valid source spends 79,904 physical actions and 86,276 input tokens, including 6,372 feedback-only tokens. Independent verification covers all 19,098 primary records, 322,100 evaluation actions and 295,280 neural decisions recomputed from checkpoints. The follow-up independently verifies 11,250 records and 185,786 policy decisions, plus 2,500 shared-prefix collection actions. Raw observation differences are zero; outcomes, rewards, identities and hash chains are reconstructed. This is reset-and-action replay, not snapshot restoration or a training reproducibility result. Fifteen boundary tests pass. No GPU or shared driver configuration is changed; training runs on CPU.

6. What Remains Unestablished

The strongest explanation currently retained is auxiliary supervision of historical facts, with direct precedent in task-inference research. It is not yet a distinctive method contribution. The symbolic reference remains at ceiling, the three learned objectives use offline teacher histories, and only

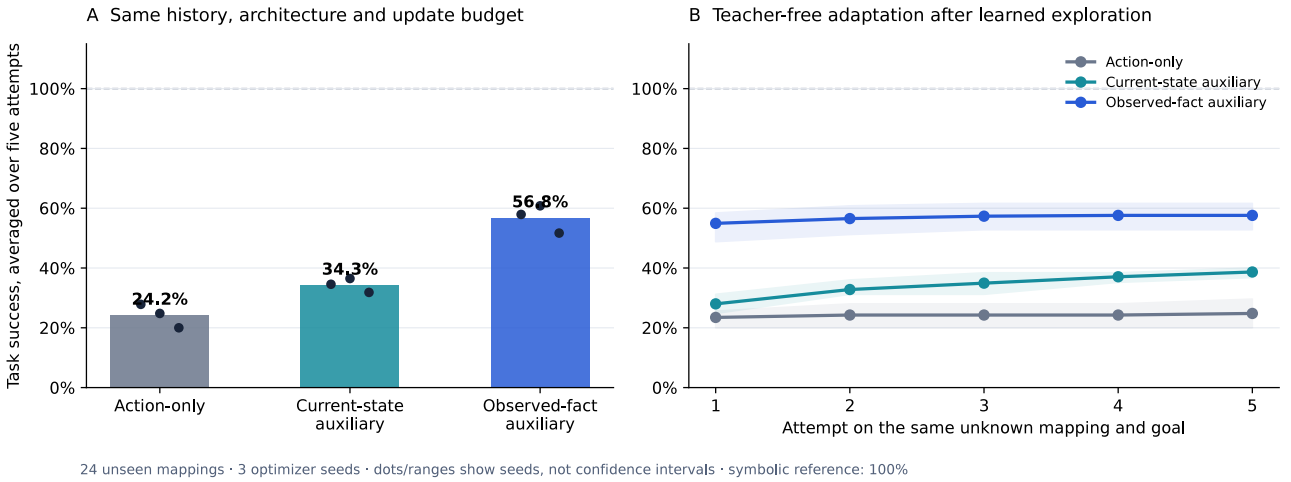


Figure 1. Matched learning budgets and complete adaptation curves. Dots and shaded ranges show the three optimizer seeds, not confidence intervals. Exploration is autonomous; the symbolic reference is a diagnostic ceiling. These results isolate auxiliary-learning behavior, not DREAM, AD or VariBAD performance.

a finite mapping family is tested. No learned exploration advantage, student-state OPD, execution-failure discrimination, future-learning-credit objective, robot transfer or recursive acceleration is established.

A meaningful next comparison must include trained task-inference and full-history methods, controlled student-history labeling, and a strong observable-belief reference. A narrower candidate concerns whether an unsuccessful action supports changing a task hypothesis when execution itself is uncertain. That would require a separately registered environment extension, matched visible feedback and explicit controls for ordinary Bayesian inference; it cannot be inferred from this deterministic Cooking pilot. Historical stopped Lift/Planar experiments remain separate.

Availability and status. Code, immutable plans, all-seed summaries, costs, verification and vector figures are at https://asimfish.github.io/seec-research/learning.html. Upstream DREAM code is not redistributed. This document is a working note in the repository's existing ICML 2026 format, not a completed method paper or submission.

References

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